

Discrete Probability distributions: Poisson and Hyper Geometric distributions

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March 25, 2026

Poisson distribution

- **Poisson distribution** is a probability distribution used when a density of items is distributed over a period of time.
- **The sample size needs to be large and the probability of success to be small.**
- Let X be a discrete r.v. It is said to follow the Poisson distribution if its pdf is of the form:

$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$ for $x = 0, 1, 2, \dots$ to infinity where λ is the parameter of the

We write : $X \sim P_0(\lambda)$

Then the Poisson distribution is defined by

$$p(k, \lambda) = p_0(\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots \text{ where } \lambda \text{ is any constant.}$$

This distribution is an infinitely countable distribution.

Example

The number of occurrences of a certain event follows a Poisson distribution with mean $\lambda = 3.5$.

- (a) Find the probability that exactly 4 occurrences happen.
- (b) Find the probability that at most 2 occurrences happen.
- (c) Find the probability that more than 1 occurrence happens.

The solution correctly applies the Poisson probability mass function:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Where $\lambda = 3.5$ **the average rate of occurrence.**

Solution

$$P(X = x) = \frac{e^{-3.5}(3.5)^x}{x!}, \quad x = 0, 1, 2, \dots$$

$$\text{a)} \quad P(X = 4) = \frac{e^{-3.5}(3.5)^4}{4!} = 0.1888 \dots$$

b)

$$\begin{aligned} P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= \frac{e^{-3.5}(3.5)^0}{0!} + \frac{e^{-3.5}(3.5)^1}{1!} + \frac{e^{-3.5}(3.5)^2}{2!} \\ &= e^{-3.5} \left(1 + 3.5 + \frac{(3.5)^2}{2!} \right) = 0.3208 \end{aligned}$$

c)

$$\begin{aligned} P(X > 1) &= 1 - P(X \leq 1) \\ &= 1 - [P(X = 0) + P(X = 1)] \\ &= 1 - (e^{-3.5} + e^{-3.5}(3.5)) = 0.8641 \end{aligned}$$

Expectation value and variance

If X is a discrete r.v such that $X \sim P_0(\lambda)$, then $E(X) = \lambda$ and $Var\ X = \lambda$

Example

If $X \sim P_0(\lambda)$ with standard deviation 3. Find

- a) $E(X)$
- b) $P(X < 4)$

Solution

a) Since standard deviation $\sigma = \sqrt{\text{Var } X}$, we have $\sqrt{\text{Var } X} = 3 \Rightarrow \text{Var } X = 9 = \lambda$

Consequently, since $X \sim P_0(\lambda)$, $E(X) = \text{Var } X = 9$

Thus the pdf of X is given by $P(X = x) = \frac{e^{-9} 9^x}{x!}$, $x = 0, 1, 2, \dots$

$$\begin{aligned} \text{b) } P(X < 4) &= P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) \\ &= e^{-9} + e^{-9} * 9 + \frac{e^{-9} * 9^2}{2!} + \frac{e^{-9} * 9^3}{3!} = 0.02122\dots \end{aligned}$$

Relationship between Binomial and Poisson Distributions

- In the binomial distribution, if n is large while the **probability** p of occurrence of an event is **close to zero**, so that $q = 1-p$ is **close to one**, the event is called a **rare event**.
- In practice, we shall consider an event as **rare** if the number of trials is at least 50 ($n \geq 50$) while np is less than 5.
- For such cases, the binomial distribution is very closely approximated by the Poisson distribution with $\lambda = np$.
- This is to be expected on comparing the equations for the means and variances for both distributions.
- By substituting $\lambda = np$, $q \approx 1$ and $p \approx 0$ into the equations for the mean and variance of a binomial distribution, we get the results for the mean and variance for the Poisson distribution. 

Hyper geometric distribution

- The hyper geometric distribution does not require independence and based on the sampling done without replacement.
- In general, we are interested in the probability of selecting x successes from the a items labeled successes and $n - x$ failures from the $N - a = b$ items labeled failures when a random of a size n is selected from N items.
- This is known as a **hyper geometric experiment**, that is, one that possesses the following two properties:
 - ① A random sample of size n is selected without replacement from N items.
 - ② x of the N items may be classified as successes and $N - x$ are classified as failures.

- The number X of successes of hyper geometric experiment is called the **hyper geometric random variable**.
- Accordingly, the probability distribution of hyper geometric variable is called the hyper geometric distribution, and its values will be denoted by $h(x, N, n, a)$.
- The total number of samples of size n chosen from N items is $\binom{N}{n}$.
- The sample are assumed to be equally likely.
- There are $\binom{a}{x}$ ways for selecting x successes for a that are available, and for each of these ways we can choose $n - x$ failures in $\binom{N-a}{n-x} = \binom{b}{n-x}$ ways.
- the total number of favorable samples among the $\binom{N}{n}$ possible samples is given by $\binom{a}{x} \binom{b}{n-x}$.

Definition

The probability distribution of the hyper geometric random variable X , the number of successes in a random of size n selected from N items of which a are labeled **successes** and $b = N - a$ labeled **failures** is

$$h(x, n, a, b) = \frac{\binom{a}{x} \binom{b}{n-x}}{\binom{a+b}{n}} = \frac{\binom{a}{x} \binom{N-a}{n-x}}{\binom{N}{n}}, \quad \max\{0, n-b\} \leq x \leq \min\{n, a\}$$

In the other words, if $n \leq a$; then $x = 0, 1, 2, \dots, n$. If $n > a$, then $x = 0, 1, 2, \dots, a$.

Example

Lots 40 components each are called unacceptable if they contain as many as 3 defectives or more. The procedure for sampling the lot is to select 5 components at a random and reject the lot. If a defective is found. What is the probability that exactly 1 defective is found in the sample if there are 3 defectives in the entire lot?

Solution: Using the hyper geometric distribution with $n = 5$, $N = 40$, $a = 3$, and $x = 1$, we have

$$h(1, 5, 3, 37) = \frac{\binom{3}{1} \binom{37}{4}}{\binom{40}{5}} = 0.030.$$

Properties

$$\begin{aligned} E(x) &= \frac{na}{a+b} = np, \\ \sigma^2 &= \frac{nab}{(a+b)^2} \frac{N-n}{N-1} = npq \frac{N-n}{N-1} \end{aligned}$$

The quantity $\frac{N-n}{a+b}$ is called the correlation factor, which is negligible when n is small relative to N .

Exercise for Binomial distribution

1

The probability that a patient recovers from a rare blood disease is 0.4. If 15 people are known to have contracted this disease, what is the probability that

- ① at least 10 survive
- ② from 3 to 8 survive
- ③ exactly 5 survive?

Exercises for poison distribution

1

During a laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond?

2

Ten is the average number of oil tankers arriving each day at a certain port. The facilities at the port can handle at most 15 tankers per day. What is the probability that on a given day tankers have to be turned away?

Exercises cont...

3

Suppose we are interested in the number of people who visit the clinic in city “X” in a given year among the total population say 5000,000, and let the probability that some one in the city visits the clinic is 0.00001. The mean number of people from the example above would be $np = 5000000 * 0.00001 = 50$ which is also the variance. For this example calculate:

- 1 The probability that no one in this population visits the clinic in the a given year
- 2 The probability that less than 5 people visits the clinic in the a given year